

GRIDGUARD AI

Predict. Prioritize. Prevent.

Power Outage Prediction & Grid Equipment Failure Advisor

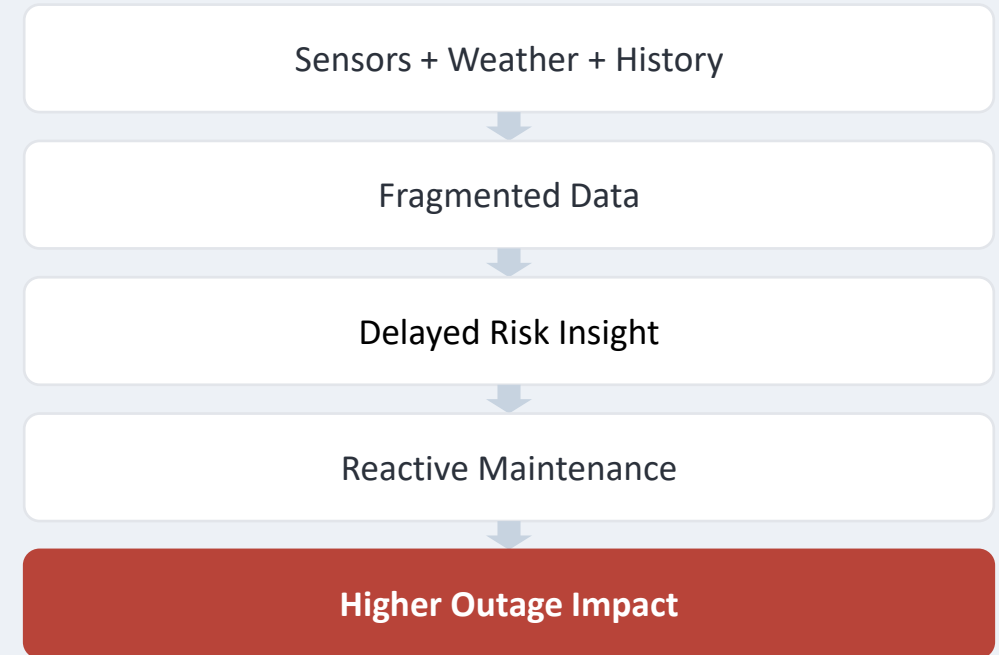
TRACK U1 — AI | IBM BoB AI INNOVATION HACKATHON 2026

TEAM GRIDGUARD

The Grid Can Warn Us — Before It Fails

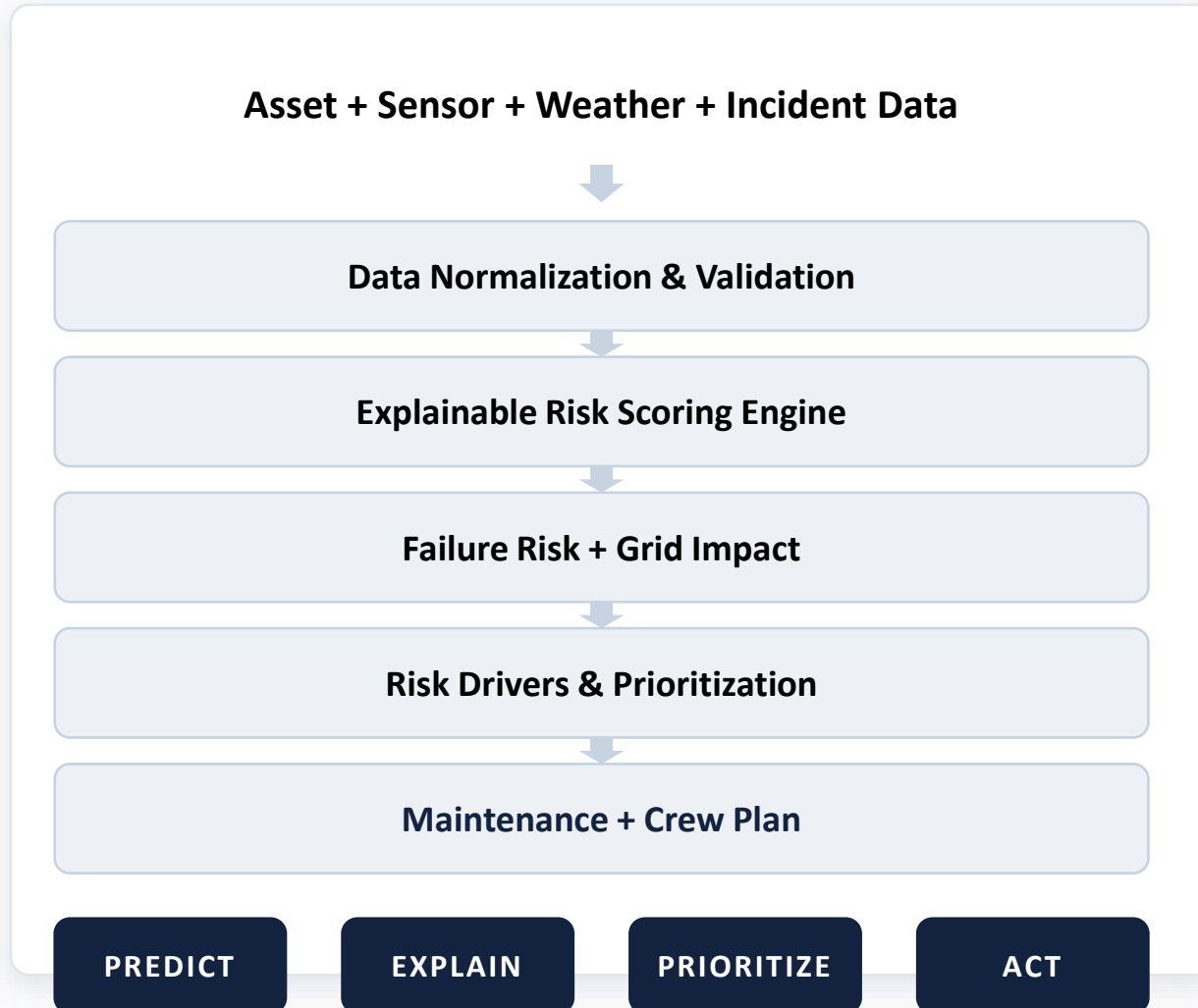
- Transformer and substation failures can trigger costly, large-scale outages.
- Sensors already capture early warning signals such as temperature, vibration, partial discharge, oil quality and load.
- Weather conditions can further increase equipment stress and outage risk.
- Sensor, weather and incident data often remain fragmented.
- Maintenance teams need to know which asset is risky, why it is risky, and what action to take first.

WHY MAINTENANCE STAYS REACTIVE



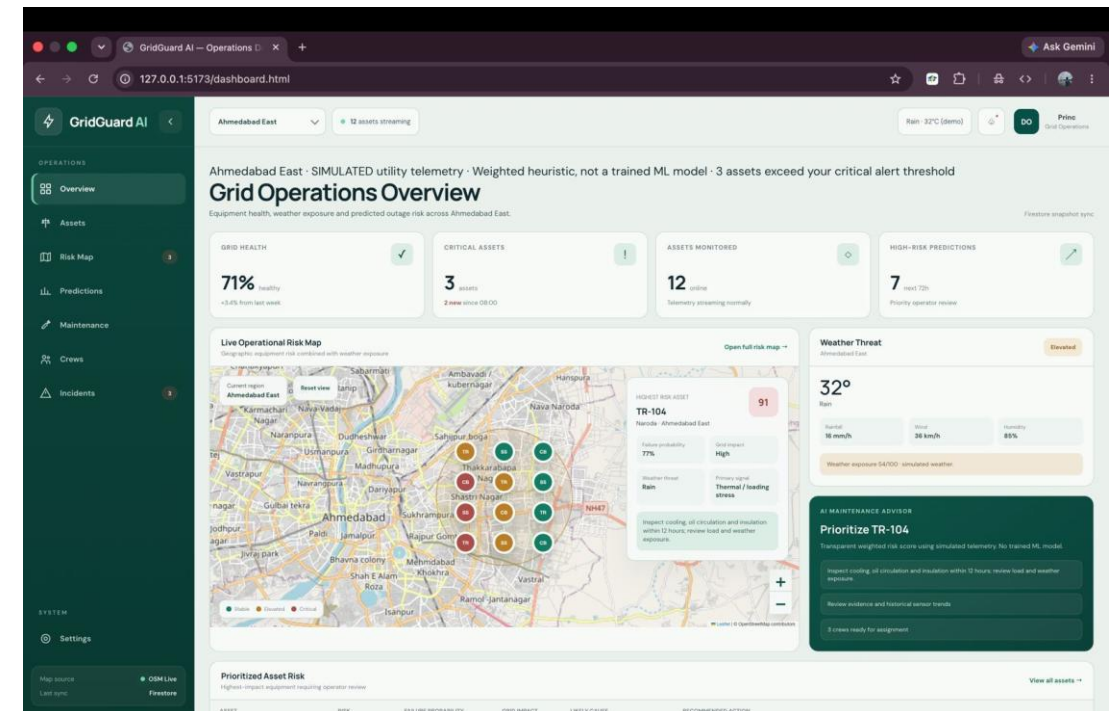
“Utilities need to know WHICH asset may fail, WHY it is risky, and WHAT to do next.”

From Grid Data to Preventive Action



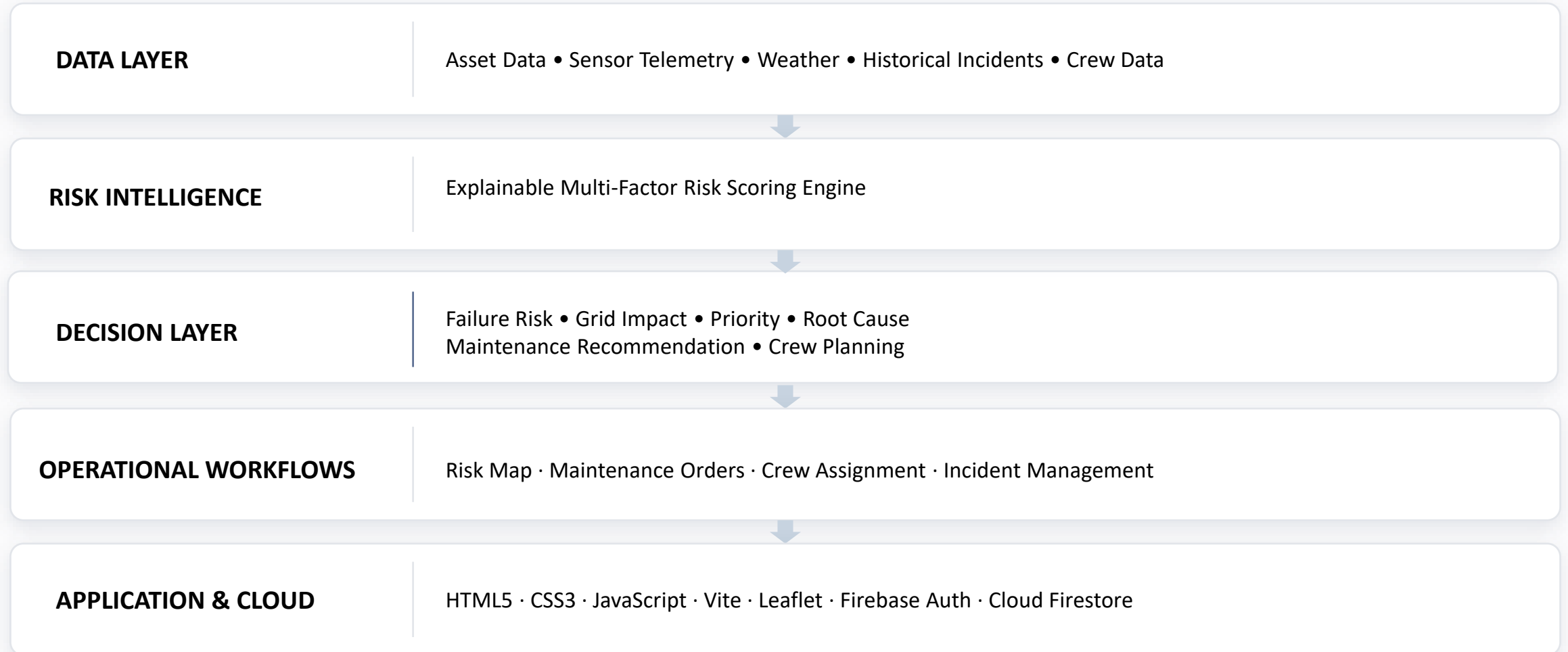
LIVE DASHBOARD PREVIEW

Stable · Elevated · Critical
Prioritized by equipment risk and grid impact



Prototype operates on simulated utility telemetry.

How GridGuard AI Works



Role-based access and Firestore Security Rules protect operational workflows.
Built as a decision-support prototype using simulated utility telemetry.

The System Doesn't Just Predict — It Explains

FAILURE RISK

Stable · Elevated · Critical

GRID IMPACT

0–100 Impact Score

PRIORITY

Risk + Grid Consequence

WHY IS IT RISKY?

- * Temperature stress
- * High electrical load
- * Vibration / condition indicators
- * Weather exposure
- * Historical incident signals

RISK INTERPRETATION

- * Identifies contributing factors
- * Highlights abnormal conditions
- * Combines multiple stress indicators
- * Keeps scoring transparent

RECOMMENDED ACTION

Prioritize immediate inspection and stage an available maintenance crew near the affected grid zone.

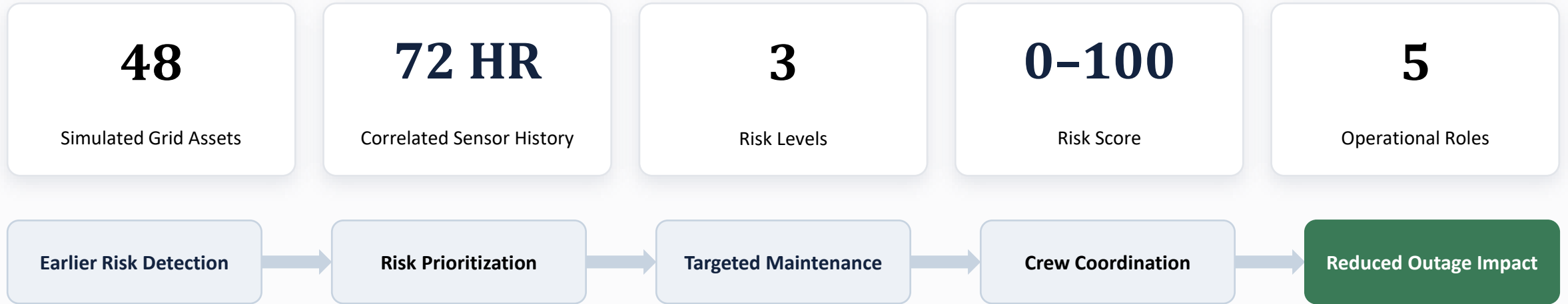
IBM BoB Powered Development

IBM BoB — AI-Assisted Development

- * Inspected and reasoned across the GridGuard codebase
- * Planned and refined application workflows
- * Assisted implementation and debugging
- * Reviewed code and validation issues
- * Supported terminal-driven testing and iteration
- * Helped accelerate documentation and engineering workflow

FRONTEND	HTML5 · CSS3 · JavaScript · Vite
CLOUD & DATABASE	Firebase · Cloud Firestore
RISK INTELLIGENCE	Explainable Multi-Factor Weighted Risk Engine
AUTH & SECURITY	Firebase Authentication · Firestore Security Rules · RBAC
MAPS	Leaflet · OpenStreetMap
TESTING & DEPLOYMENT	Node Test Runner · Playwright · Firebase Hosting

What Success Looks Like



KEY BENEFITS

Identifies high-risk grid equipment

Explains major contributing risk factors

Prioritizes risk alongside grid consequence

Converts insight into maintenance workflow

Supports crew assignment and incident response

Provides a unified geographic risk view

Prototype uses simulated telemetry and a transparent weighted risk model; real utility deployment would require calibration and validation using production grid data.

Team GridGuard

P

**Patel Prince
Ashwinkumar**

Team Lead & Primary Developer

Frontend Development · Firebase
Integration · Hosting &
Deployment · Project
Coordination

D

Patel Dwij Riteshkumar

Backend Development

Firestore Data Logic · Backend
Workflows · Risk Engine
Integration · Demo Video

H

Patel Heer Vijaykumar

Backend & UI/UX

Backend Integration ·
Operational Workflows · UI/UX ·
Demo Video

P

Patel Pal Pankajkumar

Testing & Documentation

Risk Data Analysis · System
Testing · Documentation Support
· Demo Video

“Predict. Prioritize. Prevent.”